ASBIO-C04-SIM-01 | STUDENT WORKSHEET

Osmosis and Quantitative Practical Skills

Effect of sucrose concentration on percentage change in potato-cylinder length

Name	Class	Date	Simulation Run ID

THIS ACTIVITY IS NOT GRADED

Use the HTML laboratory to prepare solutions, manipulate specimens and read instruments. Write all scientific responses, calculations, results, your graph, evaluation and reflections in this worksheet. Your teacher will collect this printed worksheet as the official record for feedback.

INDEPENDENT WORK AND METHODOLOGY

Complete the investigation independently. Do not search the internet, use AI tools, consult answer websites or ask another person to supply answers while completing it. Accessing the assigned simulation itself is permitted.

I need to see your own understanding so I can identify weaknesses and provide useful feedback. If you do not know an answer, explain what you understand or state that you are unsure. An honest incomplete answer is more useful than a copied correct answer.

Take the methodology seriously. Learn the order of the steps, why they are needed, what to measure, which variables to control and how procedural mistakes affect results. Aim to explain and reproduce the procedure independently.

MONITOR YOUR WORKING TIME

Start the actual-time stopwatch after your required reading, when you begin the investigation and worksheet. Measure the time spent on the procedure, measurements, calculations, graph and written responses. Pause it for breaks; resume when you continue. Finish it only after all required work and reflections are complete.

Your goal is not simply to finish as quickly as possible. Work accurately, independently and efficiently. Use your recorded time to identify what takes longest and where you need more fluency. Do not rush at the expense of scientific accuracy.

The page automatically pauses active time when its tab is hidden or closed. Keep it visible while working on paper; resume when you return. The 45-minute exposure clock is accelerated experimental time and is separate from your real working time.

Start time	Finish time	Break time

Total active working time: _____ hours _____ minutes _____ seconds

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BEFORE YOU BEGIN REQUIRED READING

Before opening the investigation or answering its questions, thoroughly review these topics in your AS Biology notes, Cambridge textbook and previously taught practical skills.

Biological knowledge

1. Structure and function of the cell surface membrane
2. Partially permeable membranes
3. Osmosis and water potential
4. Water movement in plant cells
5. Turgidity, flaccidity and plasmolysis
6. Effects of different external solute concentrations on plant tissues

Experimental methodology

7. Independent, dependent and controlled variables
8. Preparation of solutions using dilution calculations
9. Accurate measurement of liquid volumes
10. Standardisation of biological specimens
11. Consistent experimental procedures and fair testing
12. Recording raw measurements using appropriate precision

Quantitative practical skills

13. Calculation of percentage change
14. Construction of biological results tables
15. Graph scales, axes, units and plotting
16. Best-fit lines and identifying anomalies
17. Interpolation and interpretation of graph intercepts
18. Experimental limitations, reliability and improvements

Acknowledgement: I have reviewed these topics and understand the independent-work expectations. This is self-reported; reading is not verified by the website.

Student initials: _____

USING THE TWO RESOURCES

Follow the stage labels in this worksheet and the matching paper prompts online. Numerical syringe volumes and ruler readings may also be entered online to operate the experiment; all permanent measurements and answers belong on paper. The required graph is handwritten. The digital graph is optional practice.

Accuracy matters more than speed. This is preparation, not a competition.

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SIMULATION STAGES 01 AND 02 VARIABLES AND APPARATUS

1(a). State the independent variable.

1(b). State the dependent variable, including how it will be expressed.

1(c). State two important controlled variables. For each, explain how you will control it.

Controlled variable	How it will be controlled

Apparatus check A Additional methodology practice

At Stage 02, choose four apparatus items represented in the laboratory. Explain the purpose of each; include a specific precaution where relevant.

Apparatus	Purpose or precaution

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SIMULATION STAGE 03 DILUTION CALCULATIONS

Use 1.0 mol dm^{-3} sucrose stock and distilled water. Prepare 20 cm^3 of each treatment. Use $C_1V_1 = C_2V_2$. Label S0–S4 before adding liquid.

2. Complete the dilution table.

Treatment	Sucrose concentration / mol dm^{-3}	Stock volume / cm^3	Water volume / cm^3	Total / cm^3
S0	0.00			20
S1	0.20			20
S2	0.40			20
S3	0.60			20
S4	0.80			20

3. Show your calculation for preparing 20 cm^3 of 0.60 mol dm^{-3} sucrose from the 1.0 mol dm^{-3} stock.

Return to HTML Stage 04. Use this plan to draw, transfer and mix each solution. The syringes each hold at most 10 cm^3; repeated transfers may be necessary. Incorrect volumes are not silently corrected.

Method check B1 Why must solutions be mixed, and why should their total volumes be controlled?

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SIMULATION STAGES 04 TO 07 PREPARATION AND EXPOSURE

Record the actual stock and water volumes transferred, even if they differ from your dilution plan. Keep the target treatment identities clear.

Target treatment	Actual label	Total stock / cm³	Total water / cm³
S0 / 0.00			
S1 / 0.20			
S2 / 0.40			
S3 / 0.60			
S4 / 0.80			

Borer diameter used: _____ mm Cutting target: approximately _____ mm

Method check B2 Why should the cores come from comparable regions of the same potato?

At Stage 05, select straight, intact tissue without skin. Use one 5–6 mm borer. Wear eye protection, cut on a stable tile away from fingers, and trim close to 30 mm. At Stage 06, measure each actual initial length to the nearest 1 mm and record it immediately in the results table on the next page.

At Stage 07, fully submerge one cylinder in each solution. Keep the treatments at the same room temperature. Each cylinder receives 45 minutes of virtual exposure; the sequence uses 15-second insertion intervals.

Insertion order: _____ → _____ → _____ → _____ → _____

Method check B3 Explain how insertion and removal order keep exposure duration constant.

Method check B4 Why record raw readings immediately, and why does ruler resolution matter?

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SIMULATION STAGES 06 TO 09 MEASUREMENTS AND PROCESSING

RESULTS-TABLE CONSTRUCTION TASK

Construct one table in the ruled area below for sucrose concentration, initial length, final length, change in length and percentage change. Write all headings and units yourself. Record concentrations in order and raw lengths to the nearest 1 mm. Record each reading as soon as you observe it.

At Stage 08, remove in insertion order. For each cylinder use fresh paper towel; roll gently once; press each cut end for 5 seconds without squeezing; measure final length promptly. Record it in the same results table.

4. Explain why surface liquid must be removed using the same blotting method before final measurement.

Change in length = final length − initial length
Percentage change = (change in length ÷ initial length) × 100

5. For the 0.60 mol dm^{-3} treatment, show your calculation of change in length and percentage change. Give percentage change to one decimal place.

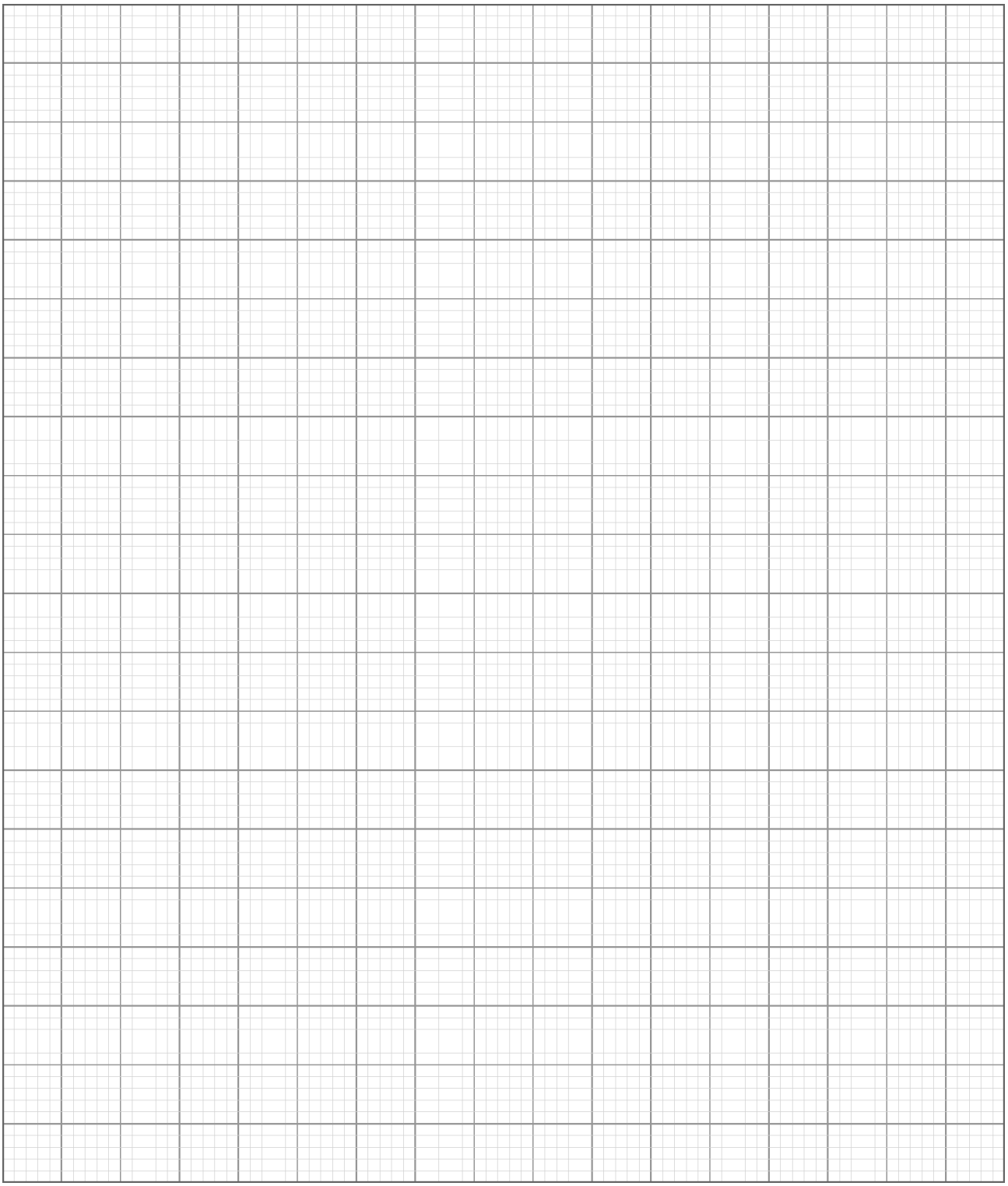
Method check B5 Why is percentage change useful when initial lengths differ?

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SIMULATION STAGE 10 HANDWRITTEN GRAPH

6. Plot a graph to show the effect of sucrose concentration on percentage change in length. Decide the appropriate axes, units and scale. Plot all valid points and draw an appropriate best-fit line or smooth curve.

Use the grid below. Decide the axes, units and a suitable linear scale yourself. Include positive and negative percentage changes and locate zero clearly. Your printed graph is required; the digital graph workspace is optional.



Use this handwritten graph for Questions 7–11 and your zero-change estimate.

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SIMULATION STAGE 11 ANALYSIS OF YOUR OWN RESULTS

7. Describe the relationship shown by your graph. Support your answer with numerical evidence.

8. Identify any anomalous result. Explain why you consider it anomalous. If there is no clear anomaly, state this and justify your decision.

9. Use your graph to estimate the sucrose concentration at which percentage change in length is 0%. State what this value means biologically.

Use your own concentrations, measurements and handwritten graph. You do not need to match another student's dataset.

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SIMULATION STAGE 11 BIOLOGICAL EXPLANATION

10. Explain the trend using water potential and osmosis. Your answer must refer to the direction of net water movement and the partially permeable cell surface membranes.

11. From your investigation, write one observation and one biological explanation. Keep the observation separate from the explanation.

Observation from the data

Biological explanation

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SIMULATION STAGE 12 EXPERIMENTAL EVALUATION

12. Complete two rows. Each row must link a specific limitation to its likely effect on the results and a practical improvement. Do not use the phrase 'human error'.

Specific limitation or source of variation	Likely effect on the results	Specific improvement

13. Explain why using at least three potato cylinders at each concentration would improve reliability, and state how the replicate results should be processed.

14. Explain why the potato cylinders must be standardised for both diameter and initial length.

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SIMULATION STAGE 12 TRANSFER TO CARROT TISSUE

15. A student investigates osmosis in carrot tissue. The student has 1.0 mol dm^{-3} sucrose stock, distilled water, a balance reading to 0.01 g and enough carrot cylinders to use three replicates at each concentration. The student will measure percentage change in mass rather than length.

15(a). State five suitable sucrose concentrations covering 0.00 to 1.00 mol dm^{-3}.

15(b). State the measurements needed and write the expression used to calculate percentage change in mass.

15(c). State two variables that should be controlled.

15(d). Explain how the graph could be used to estimate the external sucrose concentration at which there is no net movement of water.

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SIMULATION STAGE 13 PRACTICAL SKILLS READINESS CHECK

Choose one confidence level for each skill. This is self-reported and is not a grade.

1 I need substantial help.
2 I understand some parts but am still unsure.
3 I can usually do this independently.
4 I can explain and perform this confidently.

Practical skill	Circle one level
Preparing dilutions	1 2 3 4
Handling and measuring apparatus	1 2 3 4
Following the experimental sequence	1 2 3 4
Controlling variables	1 2 3 4
Recording raw measurements	1 2 3 4
Calculating percentage changes	1 2 3 4
Constructing graphs	1 2 3 4
Interpreting biological results	1 2 3 4
Explaining osmosis	1 2 3 4
Evaluating procedures	1 2 3 4
Applying skills to unfamiliar investigations	1 2 3 4

Which part of the practical methodology do you find most difficult, and why?

What practical skill will you prioritise reviewing before your assessment?

Which part of this investigation took the most time, and what could you do to work more efficiently without reducing accuracy?

Finish the stopwatch. Copy your total active time and break time into the fields on page 1. Check every question, table heading, calculation, graph and reflection. Submit this printed worksheet to your teacher. No digital export is required.

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